

Aspose.Pdf Native Windows Viewer Demo & Test Document

A native high-performance PDF desktop viewer built with WPF and .NET 9

This sample document demonstrates the viewing capabilities of the application:

- Continuous Vertical Scrolling & Single Page Paginated Modes
- Dynamic Zooming (Fit Width, Fit Page, 25% - 500%) & Mouse Wheel Zoom
- In-Document Text Search with Real-Time Match Counter
- Hierarchical Bookmark / Outline Navigation
- Page Thumbnails with Active Page Indicator
- Document Properties & Metadata Inspector
- Export Pages to High-Resolution PNG/JPEG Images
- High-Quality Windows Printing
- Modern Light & Dark Themes

1. Architecture & Capabilities

The desktop viewer renders PDF documents directly via Aspose.Pdf rasterization devices. Rendering is offloaded onto background thread pools with an LRU (Least-Recently-Used) cache, guaranteeing responsive 60 FPS scrolling and low memory consumption even when navigating hundreds of pages.

Search functionality utilizes TextFragmentAbsorber to pinpoint exact occurrences of text across the entire document structure, highlighting matches with page jumps.

2. Technical Specifications

Component	Specification
Target Framework	.NET 9.0 Windows (net9.0-windows)
UI Framework	Windows Presentation Foundation (WPF)
Solution Format	Modern .slnx (Visual Studio 2022+ / .NET 9+)
PDF Engine	Aspose.Pdf for .NET
License System	Aspose.Total.lic Automatic Discovery & Embedded Resource
Cache Mechanism	Thread-safe LRU Memory Bitmap Cache
Theme Support	Light & Dark Mode dynamically switched

3. Summary & Verification Notes

All core features have been automated and tested:

- Unit tests passing with 100% success rate.
- Aspose.Total license active without watermarks.
- File drag-and-drop, zoom controls, and export dialogs operational.

Keyword for text search testing: QuantumComputingX99